

SALMA EL MAMOUNI

AI Engineer - Machine Learning, NLP, Deep Learning Systems

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PROFILE

AI Engineer with hands-on experience building and deploying Machine Learning, Deep Learning and NLP systems end-to-end, from data pipelines and model training to evaluation and API-based delivery. Master's degree in Artificial Intelligence and Digital Computing, combining strong ML/DL foundations with software engineering skills (Java, Spring Boot, React, REST APIs) to ship production-ready AI solutions. Seeking a first full-time role as an AI Engineer or ML Engineer.

EXPERIENCE

Full-Stack and API Development Intern - ALTEN Maroc, Fes (2024, 2 months)

- Designed and built a secure web application to centralize visitor registration, tracking and management, from requirements analysis to deployment.
- Built REST APIs with Spring Boot and React front-end interfaces; implemented authentication, role-based access control and MySQL data persistence, engineering skills directly transferable to serving and integrating ML models in production.

Tools: Java, Spring Boot, Spring Security, React.js, MySQL, Axios, REST API, Postman

AI AND ML ENGINEERING PROJECTS

Applied AI for Solar Energy Forecasting - Capstone Project - FST Beni Mellal (2026)

- Built an end-to-end NLP pipeline processing a multilingual corpus (French, Arabic, English, Darija) from 25,938 news articles and 604,257 web comments, cleaned and deduplicated into 16,642 articles and 114,560 usable comments; applied XLM-RoBERTa, BERTopic and zero-shot classification for sentiment and topic analysis.
- Designed, trained and benchmarked 6 hybrid Transformer-LSTM/GRU/BiLSTM deep learning architectures for photovoltaic output forecasting across 3 Moroccan solar sites (NOOR Ouarzazate, Midelt, Boujdour), reaching up to 95.9% accuracy (10% tolerance) with the RNN-Transformer-BiLSTM model.
- Evaluated cross-site model generalization on over 166,000 hourly observations per site, validating robustness for real-world deployment across different environments.

Tools: Python, REST APIs, Pandas, NumPy, TensorFlow, Keras, Scikit-learn, XLM-RoBERTa, BERTopic

Heart Disease Prediction - Supervised Classification Pipeline

- Built a full ML pipeline: KNN-based missing value imputation, feature standardization, feature selection (Relief, LASSO), then trained and compared Random Forest, KNN and AdaBoost classifiers to identify the best-performing model.

Tools: Python, Scikit-learn, Random Forest, KNN, AdaBoost

Real-Time News Monitoring System - Streaming and Graph Data

- Built a real-time data pipeline with Kafka for continuous ingestion, modeled entity relationships in Neo4j, and delivered an interactive Streamlit dashboard for live monitoring.

Tools: Python, Kafka, Neo4j, Cypher, Streamlit

Business Intelligence Solution - OLAP and Dashboards

- Designed an OLAP cube in SSAS, queried it with MDX, and built interactive Power BI dashboards for multidimensional analysis.

Tools: SSAS, OLAP, MDX, Power BI

EDUCATION

Master's Degree, AI and Digital Computing - FST Beni Mellal, USMS (2024-2026)

Machine Learning, Deep Learning, NLP, Computer Vision, Reinforcement Learning, Time Series, Big Data and Cloud Computing

Bachelor's Degree in Computer Science - FST Beni Mellal (2023-2024)

DEUST MIPC - FST Beni Mellal (2021-2023)

Baccalaureate, Physical Sciences - Beni Mellal (2021)

TECHNICAL SKILLS

ML and Deployment	Classification, regression, preprocessing, feature engineering and selection, model training, evaluation and comparison
Deep Learning	Neural networks, RNN, LSTM, GRU, BiLSTM, Transformer architectures
NLP	Sentiment analysis, topic modeling, zero-shot classification, XLM-RoBERTa, BERTopic
Time Series	Sequence preparation, forecasting, performance evaluation, generalization
Programming	Python, Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Matplotlib
Software Engineering	Java, Spring Boot, Spring Security, React.js, REST APIs, Git
Data and Big Data	SQL, MySQL, MongoDB, Neo4j, Kafka, Hadoop, HDFS, Spark, Hive, YARN
BI and Analytics	Power BI, SSAS, OLAP, MDX

LANGUAGES

Arabic (Native) | French (Good level) | English (Technical level)

STRENGTHS

Team work, Leadership, Time management, Critical thinking, Fast learner